

Additive Build Advisor — Technical Report

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1 Technical Report: Additive Build Advisor

1.1 Overview

The Additive Build Advisor is a personal project I built to learn the **design-to-inspection digital thread** for additive manufacturing end to end, and to have a clean, extensible base I can keep developing. It ingests a part as an STL mesh, recovers a watertight geometry, chooses a build orientation by resting the part on candidate flat faces, simulates the build on a voxel model, solves a finite-element **distortion analysis** with the inherent-strain method, checks manufacturability (DfAM), turns the part's tolerances into an inspection plan, and assembles a single machine-readable record with an explicit release gate.

The aim is not a production build processor. It is to implement the whole workflow myself, from first principles where that aids understanding and with an established library where that is the right tool, so each step is transparent and I can extend any piece later.

The STL parser, geometry kernel, voxelizer, orientation search, and build simulation are written from scratch on **numpy**. The distortion FEA is assembled and solved with **scikit-fem** (on **scipy**);

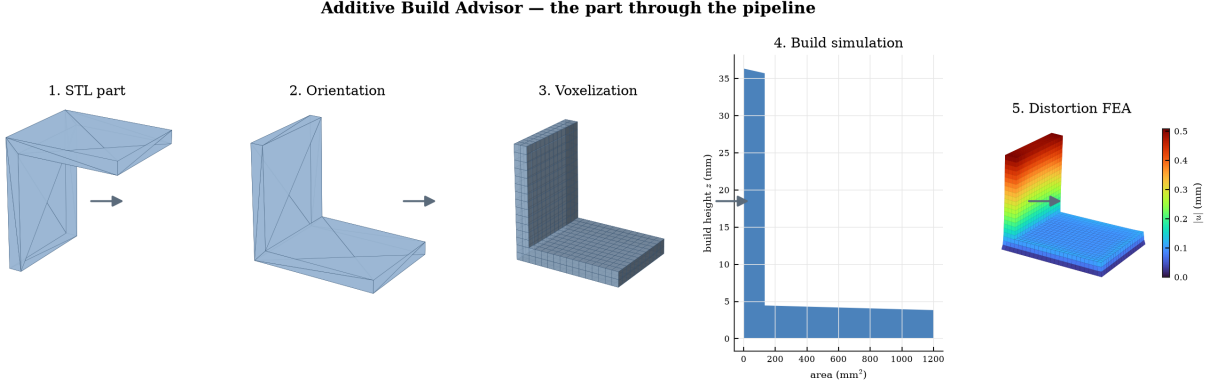


Figure 1: The part through the pipeline

`matplotlib` renders the figures.

1.2 System architecture

The pipeline is: geometry recovery $\rightarrow$ orientation screening $\rightarrow$ voxelization $\rightarrow$ three analyses on the voxel model (build simulation, distortion FEA, DfAM) plus an inspection plan from the tolerances $\rightarrow$ a release gate $\rightarrow$ a digital-thread record that can hand off to a runtime monitoring twin.

1.3 Geometry and the voxelization engine

The STL is parsed (binary and ASCII), normals are recomputed from vertex winding rather than trusted from the file, and the mesh volume is computed analytically from the divergence theorem as a sum of signed tetrahedra over the F facets,

$$V = \frac{1}{6} \left| \sum_{f=1}^F \mathbf{a}_f \cdot (\mathbf{b}_f \times \mathbf{c}_f) \right|,$$

where $\mathbf{a}_f, \mathbf{b}_f, \mathbf{c}_f$ are the vertices of facet f . A watertight/manifold check welds coincident vertices and confirms every edge is shared by exactly two facets.

A triangle soup is not a solid model, so the advisor builds an occupancy grid by *ray stabbing*: for each (x, y) column it shoots a vertical ray, collects the z heights where the ray crosses the mesh, sorts them, and fills voxels between entry/exit pairs (the even-odd / crossing-number rule — a point is inside when a ray to infinity crosses the surface an odd number of times). Per-axis grid jitter keeps sample points off face diagonals, coincident crossings are de-duplicated so the pairing stays correct on shared edges, and z is sampled at voxel centres so the discretised volume is unbiased. The grid drives the support, thin-wall, trapped-void and per-layer cross-section estimates, and supplies the element mesh for the FEA.

1.3.1 Engine validation

The voxel/mesh volume error is reported on every run. For an axis-aligned cube (10 mm, analytic 1000 mm^3) the discretisation is exact:

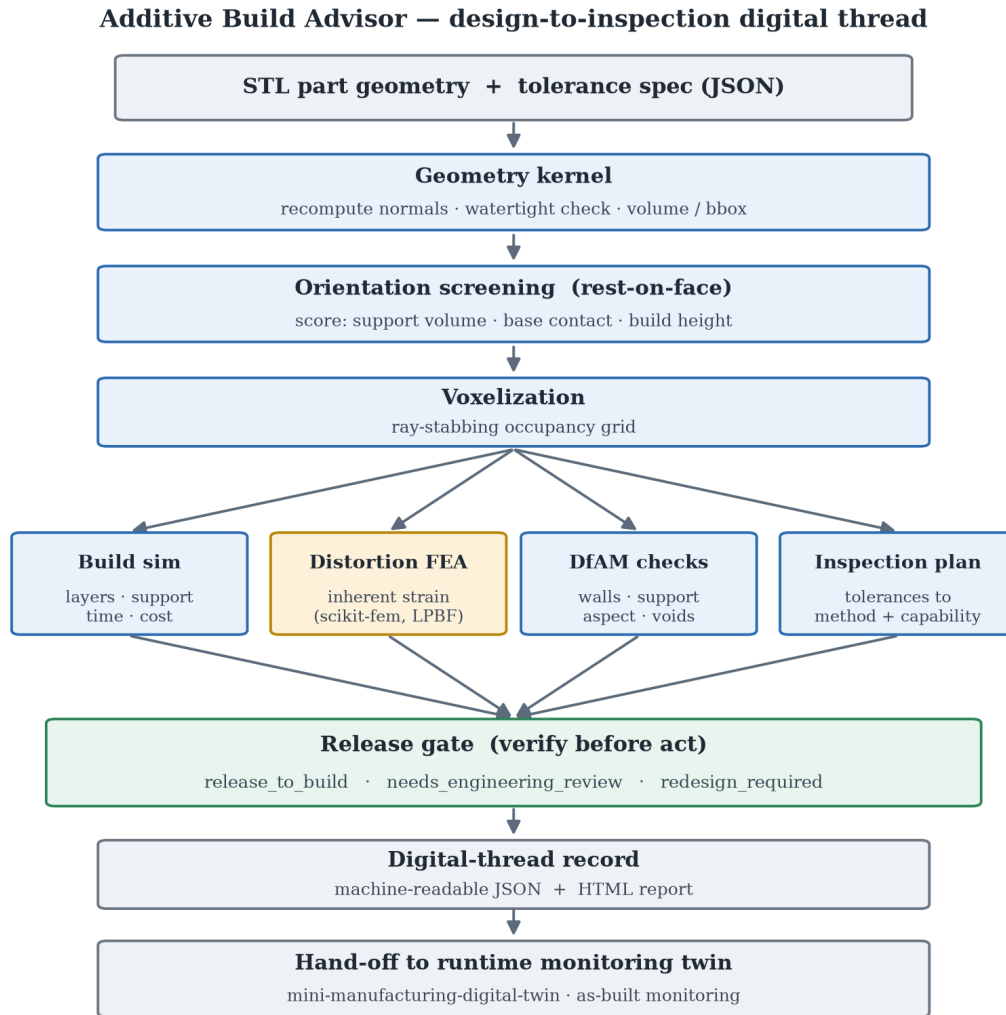


Figure 2: System diagram

grid_n	voxel volume (mm ³)	error
16	1000.00	0.00%
32	1000.00	0.00%
64	1000.00	0.00%

For an off-axis rotated bracket (analytic 8640 mm³) it converges as the grid refines, staying under ~0.4% even when coarse (+0.18%, +0.03%, -0.02% for grid_n 24, 48, 96). A solid cube reports zero trapped volume and zero support; a hollow box with a 216 mm³ internal void recovers ~220 mm³ by flood fill.

1.4 Orientation: rest on a flat face

Orientation is the highest-leverage additive decision. Candidates are generated the way a machinist chooses — *which face goes down* — by clustering the mesh’s facet normals into its significant flat faces and adding the six bounding-box directions as a fallback, so every candidate rests a real face on the plate.

Each candidate is screened on a coarse voxelization and scored by a normalised, weighted objective (lower is better),

$$\text{score} = w_S \tilde{S} + w_c (1 - \tilde{c}) + w_H \tilde{H} + P, \quad (w_S, w_c, w_H) = (0.45, 0.35, 0.20),$$

where $\tilde{S}$ is the min-max normalised **support material volume**, $\tilde{c}$ the **base-contact fraction** (footprint actually touching the plate), $\tilde{H}$ the **build height**, and P a hard penalty for orientations that exceed the build volume. On the sample bracket the winner rests on the large flat back face — full base contact and zero support — versus alternatives that need more than 5 cm³ of support on a tiny contact patch.

A note on a pitfall I hit: an earlier version scored orientation by overhang *facet area* alone and chose a 45° edge-balanced tilt. It drove the overhang metric to zero but was physically wrong — no flat base and large support. Scoring the quantity that actually costs (support volume) plus base contact fixed it. A metric that is cheap to game is the wrong metric.

1.5 Build simulation

From the oriented mesh and grid, the simulation estimates the layer count and build time,

$$N_{\text{layers}} = \left\lceil \frac{H}{h} \right\rceil, \quad t = \frac{V_{\text{part}} + V_{\text{support}}}{\dot{V}} + N_{\text{layers}} t_{\text{recoat}},$$

with H the build height, h the layer height, $\dot{V}$ the effective deposition/fusion rate, and t_{recoat} the per-layer overhead; cost is material mass times unit cost plus machine time times machine rate. Support material is the support infill fraction times the empty volume that sits beneath solid. The per-layer cross-section comes straight from the voxel grid:

These scale correctly across processes: the same bracket is ~160 layers and a few dollars on FFF but ~1000 layers and a few hundred dollars on metal LPBF, driven by the 0.03 mm layer height and the machine rate.

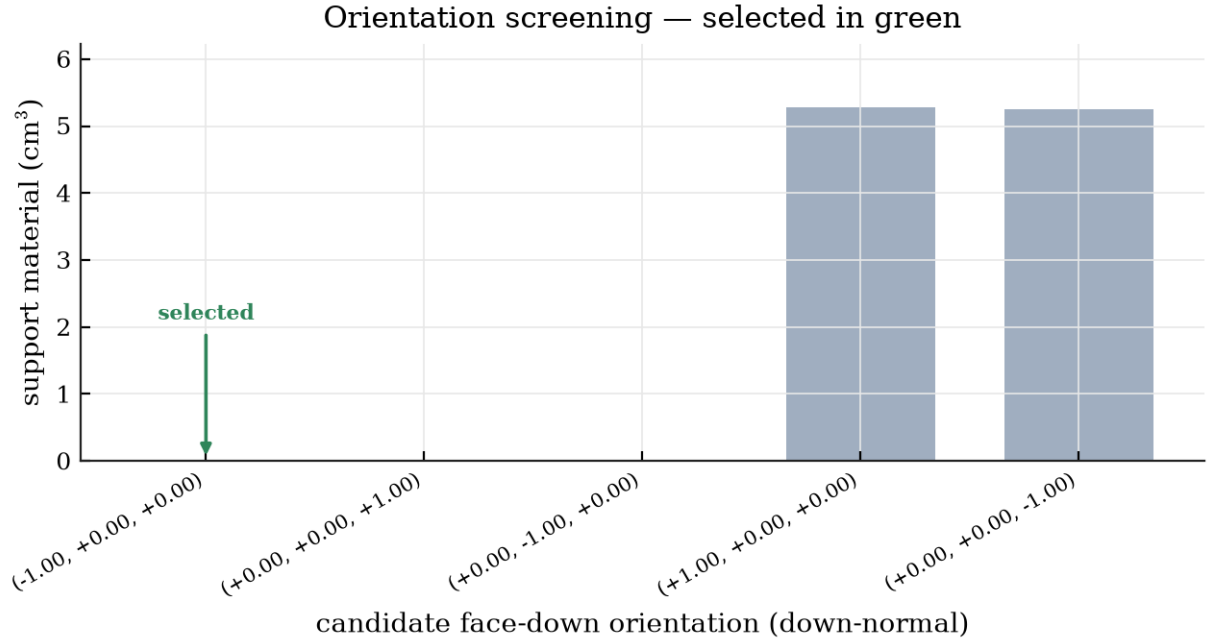


Figure 3: Orientation screening

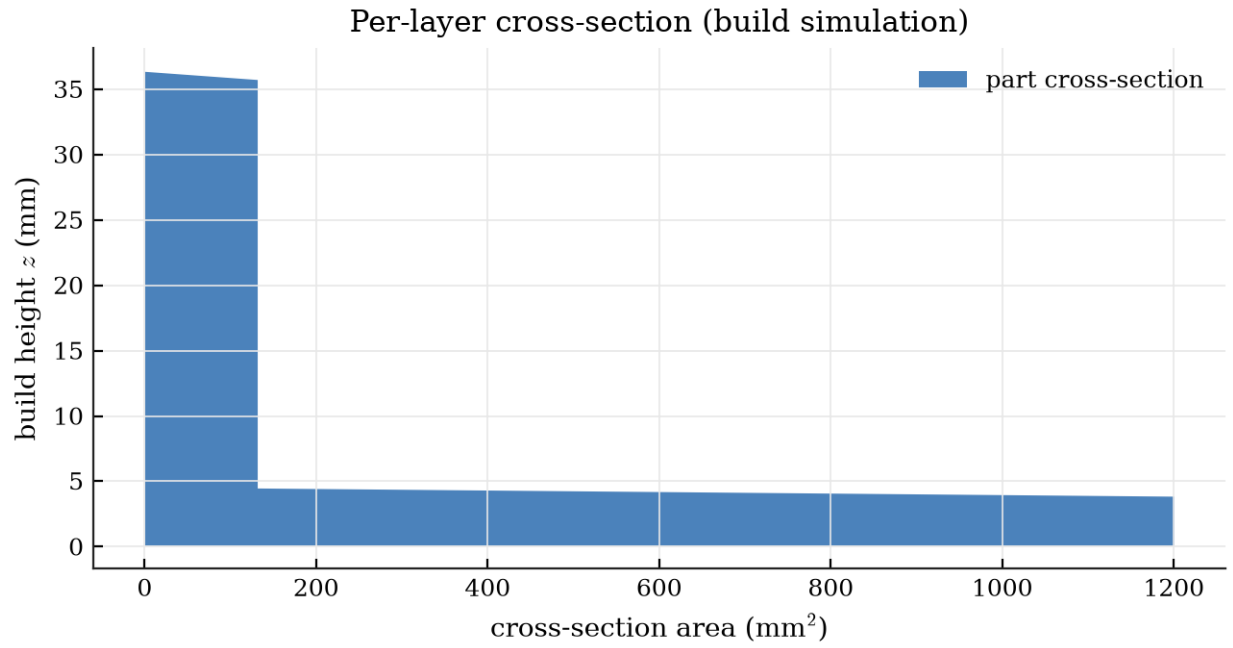


Figure 4: Per-layer cross-section

1.6 Distortion FEA: the inherent-strain method

Distortion is predicted with a genuine finite-element solve, assembled and solved with **scikit-fem** on a hexahedral mesh of 8-node trilinear elements (good for bending, unlike linear tetrahedra). The constitutive law carries an eigenstrain ε^* (the inherent strain that models the build’s accumulated thermal shrinkage),

$$\sigma = \mathbf{C} : (\varepsilon - \varepsilon^*), \quad \lambda = \frac{E\nu}{(1+\nu)(1-2\nu)}, \quad \mu = \frac{E}{2(1+\nu)}.$$

The weak form is: find $\mathbf{u}$ such that for all test fields $\mathbf{v}$,

$$\int_{\Omega} \varepsilon(\mathbf{v}) : \mathbf{C} : \varepsilon(\mathbf{u}) dV = \int_{\Omega} \varepsilon(\mathbf{v}) : \mathbf{C} : \varepsilon^* dV.$$

For an isotropic eigenstrain $\varepsilon^* = \varepsilon^* \mathbf{I}$ the eigenstress is hydrostatic and the consistent load reduces to a divergence term,

$$\sigma_0 = (3\lambda + 2\mu) \varepsilon^*, \quad \mathbf{f} = \int_{\Omega} \sigma_0 \nabla \cdot \mathbf{v} dV,$$

giving the discrete system $\mathbf{K}\mathbf{u} = \mathbf{f}$, with the base layer clamped to the plate ($\mathbf{u} = \mathbf{0}$ at $z=0$) and the sparse solve performed by SciPy. The displacement field is the predicted distortion; clamping the base while the bulk shrinks reproduces the corner-lift that dominates real additive distortion. The element von Mises stress,

$$\sigma_{vM} = \sqrt{\frac{1}{2} [(\sigma_x - \sigma_y)^2 + (\sigma_y - \sigma_z)^2 + (\sigma_z - \sigma_x)^2] + 3(\tau_{xy}^2 + \tau_{yz}^2 + \tau_{zx}^2)},$$

is recovered for context. The result is drawn as a deformed element mesh (exaggerated for visibility), contour-coloured by displacement magnitude:

A useful property: for an eigenstrain-only load (no external force) both $\mathbf{K}$ and $\mathbf{f}$ scale linearly with E , so the displacement $\mathbf{u} = \mathbf{K}^{-1}\mathbf{f}$ is **independent of Young’s modulus** — distortion is governed by geometry, eigenstrain and Poisson’s ratio. E only enters the stress. That peak stress is linear-elastic and *indicative* only: with no plasticity it can exceed yield, where a real part would yield and stress-relieve. The distortion field is the meaningful output.

1.6.1 Target process, basis, and prior art

This targets **metal laser powder-bed fusion (LPBF)**, where residual-stress warpage is the governing failure mode. The inherent-strain method — calibrate an eigenstrain (from Ueda’s inherent-strain theory, adapted to AM) and apply it as a static load to a part-scale elastic FEA — is the accepted part-scale approach and what commercial tools (Netfabb, ANSYS Additive, Simufact) implement; see the state-of-the-art review in *Int. J. Adv. Manuf. Technol.* (2022). The recognised validation artifact is the **NIST AM-Bench 2018 (AMB2018-01)** single cantilever / 12-leg bridge, measured by CMM before and after EDM release. The repo includes the cantilever geometry and an IN625 profile to run on it. For polymer processes the same solver runs, but the record marks the result *indicative only*, since the eigenstrain calibration is a metal-PBF notion.

Inherent-strain FEA — distortion on deformed mesh (deformation $\times 10$)
peak 0.311 mm, 2820 hex elements, von Mises peak 45 MPa

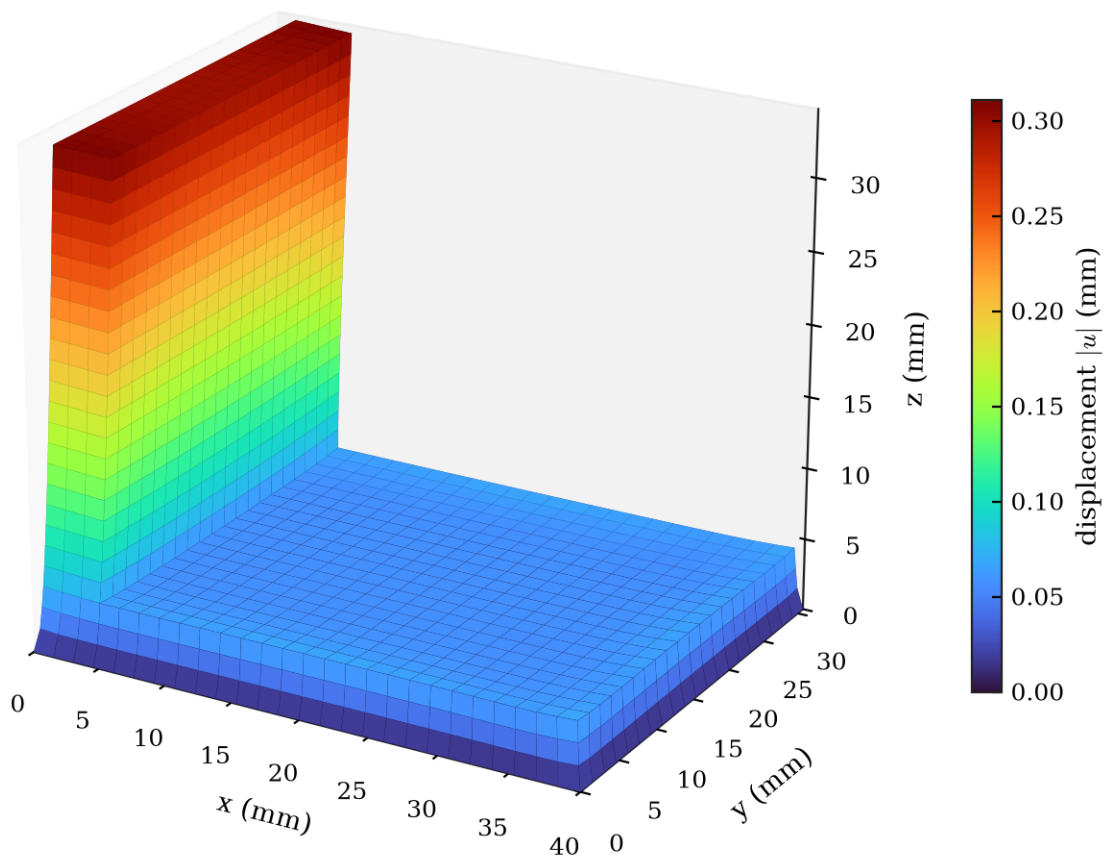


Figure 5: Inherent-strain FEA distortion

1.6.2 What the number means: on-plate vs post-release

This model holds the part **bonded to the plate** and reports the *on-plate* distortion field. The NIST cantilever’s headline $\sim 1.0\text{--}1.3\text{ mm}$ is the **post-release** deflection measured after the part is cut from the plate, when stored residual stress relaxes into a large curl — a different, larger quantity. The predicted on-plate cantilever distortion ($\sim 0.1\text{ mm}$) is therefore not directly comparable to that 1.3 mm , and the tool says so rather than forcing the match. Capturing the post-release deflection needs a build-plate release/cutting step and a calibrated (anisotropic, layer-activated) inherent strain — see “Where I’d take it next.”

1.6.3 FEA validation

A prismatic bar clamped at the base under a uniform eigenstrain has an analytical top displacement $u_{\text{top}} = |\varepsilon^*| H$. Refining the mesh, the FEA converges toward it:

voxel pitch (mm)	FEA peak distortion (mm)	error
4.0	0.4213	+5.3%
2.0	0.4161	+4.0%
1.0	0.4141	+3.5%
0.5	0.4132	+3.3%

(analytic axial value 0.400 mm for $\varepsilon^* = -0.01$, $H = 40\text{ mm}$). The residual few percent is the lateral corner motion: the reported metric is the peak displacement *magnitude*, which at the top corner adds in-plane shrinkage to the axial component; the axial component itself converges to 0.400 mm . Running the same bracket across every process confirms the displacement is linear in the eigenstrain and independent of E — exactly what linear elasticity requires for an eigenstrain-only load:

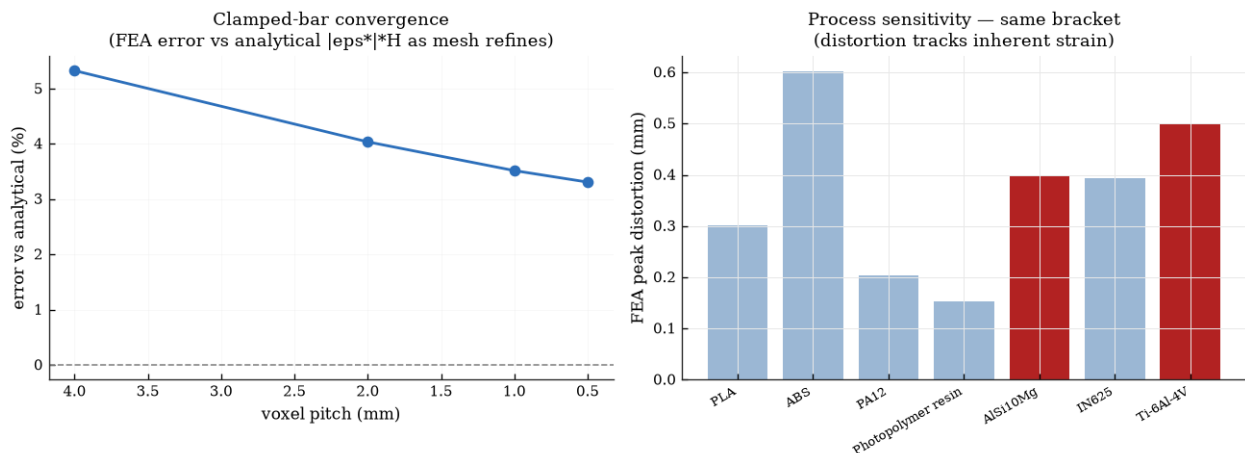


Figure 6: FEA validation

1.7 Cross-process comparison

The build simulation, cost/time, and DfAM checks run natively for every process profile. The distortion FEA is grounded in metal LPBF (the inherent-strain method is a metal-PBF technique)

and is shown for FFF and SLA as an *indicative* comparison. Running the same bracket through three processes:

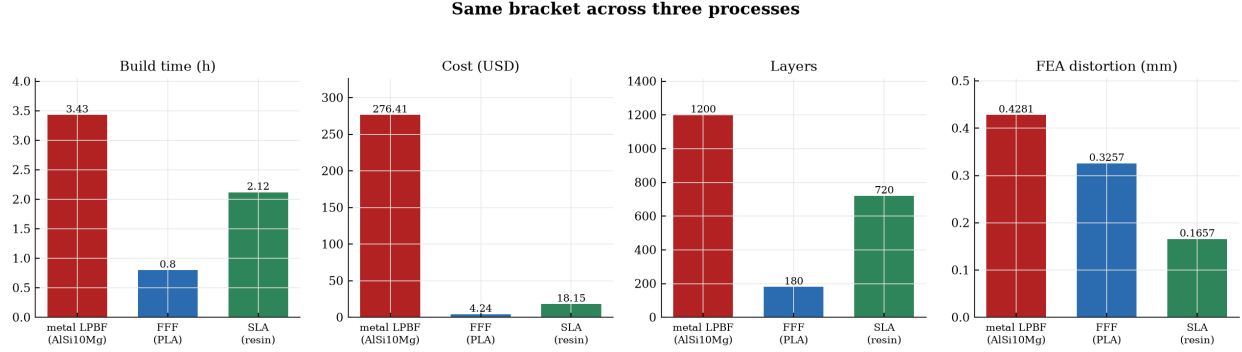


Figure 7: Cross-process comparison

Process	Build time	Cost	Layers	FEA distortion
metal LPBF (AlSi10Mg)	3.43 h	\$276	1200	0.428 mm
FFF (PLA)	0.80 h	\$4.24	180	0.326 mm
SLA (resin)	2.12 h	\$18.15	720	0.166 mm

Metal is the slowest, most expensive, and highest-distortion route; FFF is fastest and cheapest; SLA sits between, with the finest layers among the polymers. The distortion ordering follows each process’s representative inherent strain (and is E -independent), so the FFF/SLA values indicate relative shrinkage tendency rather than calibrated predictions.

1.8 DfAM checks

Transparent, severity-ranked manufacturability rules read from the same grid, simulation, and FEA as the headline numbers:

Check	Method
Build-volume fit	bounding box vs machine envelope
Thin walls	morphological opening at the min-wall radius
Support burden	support material as a fraction of part volume
Aspect ratio	height / min footprint dimension
Trapped volume	flood fill from outside; unreachable voids are enclosed
Distortion	FEA peak displacement $\delta_{\max}$ as a fraction $r = \delta_{\max} / \max(\ell_x, \ell_y, \ell_z)$

1.9 Inspection planning and process capability

The planner turns each toleranced dimension, GD&T control, and surface-finish requirement into an inspection step, choosing a method by how tight the tolerance is (calipers $\rightarrow$ micrometer $\rightarrow$ CMM $\rightarrow$ CT) and checking each tolerance against the process’s **as-built capability**. A tolerance

below as-built capability is flagged as needing post-machining, because inspecting to an impossible tolerance only confirms a guaranteed failure. The tolerance spec is plain JSON, so the design data stays CAD-neutral (a Fusion or STEP exporter would populate the same fields).

1.10 The release gate

The advisor never silently approves a build. It returns one of three decisions with reasons and blocking findings attached:

Decision	When
<code>release_to_build</code>	DfAM clean, tolerances within capability, simulation validated
<code>needs_engineering_review</code>	warnings, tolerances needing post-machining, or low simulation confidence (unsealed mesh / coarse-grid volume error)
<code>redesign_required</code>	a critical DfAM finding (does not fit, traps resin/powder, severe thin walls)

A model may recommend, but a physical action is gated on evidence, on the confidence of the model, and on a human-review path when anything is uncertain. The record carries an explicit hand-off block so design intent can flow into as-built monitoring. Worked outcomes from the sample run: `calibration_cube` / **FFF** → `release_to_build`; `gantry_bracket` / **FFF** → `needs_engineering_review` (± 0.05 mm and $3.2\text{ }\mu\text{m}$ are below FFF capability → route to machining); `hollow_housing` / **SLA** → `redesign_required` (enclosed cavity traps resin → add drain holes).

1.11 Validation summary

```
pytest # 10 tests
python examples/validate_fea.py # regenerates the FEA validation figure
```

The tests cover: sample parts are watertight; STL round-trips; the cube discretises exactly; an off-axis part's volume converges; trapped volume is detected; the chosen orientation rests on a real face and is not the highest-support candidate; the FEA matches the analytical clamped-bar solution within 10%; distortion appears in the record; all three gate outcomes fire; and the record is JSON-clean.

1.12 Limitations

Stated plainly, because the value here is the workflow and the engineering, not production fidelity. The tool does **not** include:

- A true slicer / toolpath generator (build time is volume- and layer-based).
- A calibrated transient thermo-mechanical solve. The FEA is a real linear-elastic solve, but the inherent strain is a representative per-process value, not fit to melt-pool thermal history.
- OEM-qualified material/machine profiles (numbers are representative defaults).
- A real CAD/CAM integration (geometry is STL; tolerances are JSON).
- Lattice/infill modeling, multi-part nesting, or layer-by-layer activation in the FEA (the eigen-strain is applied to the whole part at once).

- Validation against measured build outcomes.

1.13 Where I'd take it next

1. A real slicer with toolpath-length-based time and proper support generation.
2. Layer-by-layer element activation and a calibrated inherent-strain (or transient thermo-mechanical) solve, validated against measured distortion, including a build-plate release step so post-release deflection can be compared to NIST.
3. Qualified per-machine process profiles and measured-vs-predicted calibration.
4. CAD/CAM integration (Fusion / STEP) to pull features, datums and PMI directly into the same record schema.
5. Closed-loop feedback: as-built inspection and in-situ monitoring updating the capability data and the eigenstrain calibration.
6. A persisted thread store linking design revision $\rightarrow$ build $\rightarrow$ inspection $\rightarrow$ part.